Abdul Rahiman Mihad

AI Engineer | LLM Agents, RAG, Real-Time Voice AI, MLOps

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Nationality: Indian | **Visa Status:** UAE Visit Visa | **Availability:** Immediate Joiner

PROFESSIONAL SUMMARY

AI Engineer who ships production LLM systems end to end. At Vendoor, a B2B tender platform for the UAE food and beverage market, built **four production LLM agents** with enforced structured output, semantic catalog matching on pgvector, WhatsApp and email automation, and a FastAPI and PostgreSQL backend on AWS. Previously built Alpha, a full-duplex real-time AI voice interviewer at HireUp, lifting navigation accuracy from **about 80% to 100%** and cutting startup latency by **about 1.5 seconds**. Strong in Generative AI, agentic systems, RAG, NLP, Computer Vision and MLOps. Microsoft Azure certified (AI-900). Based in the UAE and available to join immediately.

CORE COMPETENCIES

Generative AI | Large Language Models (LLMs) | AI Agents | Agentic AI | Multi-Agent Systems | Tool Use and Structured Output | Retrieval-Augmented Generation (RAG) | Vector Search | Embeddings | Prompt Engineering | LLM Evaluation | Conversational AI | Voice AI | Real-Time Streaming | Natural Language Processing (NLP) | Computer Vision | LLM Fine-Tuning (PEFT/LoRA) | LangGraph | LangChain | Model Context Protocol (MCP) | Python | FastAPI | AWS | Docker | CI/CD | MLOps

WORK EXPERIENCE

Full-Stack AI Engineer, Vendoor **Jul 2026 to Present**

B2B supplier discovery and tender platform for the UAE food and beverage market, UAE (early-stage startup)

- Shipped **4 production LLM agents** with enforced structured output that turn buyer specs, spreadsheets and **5,000-SKU** supplier catalogs into tender line items, and parse free-text supplier quotes with per-line confidence scores.
- Built semantic catalog matching with Voyage AI embeddings on pgvector and a pricing normalisation engine, making every supplier price comparable at the buyer's quantity and matching synonyms such as aubergine and eggplant.
- Automated supplier outreach and quote collection over the WhatsApp Cloud API and email, with HMAC-verified webhooks and atomic ingestion so retries never duplicate a quote and no message waits on a human.
- Ported all **86 API routes** with a **zero-downtime** cut-over while migrating the backend from Node.js and MySQL to Python 3.12, FastAPI and PostgreSQL on AWS (App Runner, ECS Fargate, RDS, S3, SQS), under a CI-enforced contract.

Tech: Python 3.12, FastAPI, PostgreSQL, pgvector, Voyage AI, SQLAlchemy, Alembic, Anthropic Claude API, WhatsApp Cloud API, AWS, GitHub Actions, pytest

AI/ML Engineer, HireUp **Jan 2026 to Jun 2026**

Technology company headquartered in Japan (remote). Built Alpha, a real-time AI voice interviewer, app.hireup.now

- Raised interview navigation accuracy from **about 80% to 100%** by replacing non-deterministic LLM routing with a deterministic resolver, verified across live interviews and automated tests.
- Cut startup latency by **about 1.5 seconds** by designing the full-duplex real-time voice pipeline: WebSocket streaming, concurrent asyncio loops, Deepgram Flux speech-to-text, WebRTC VAD and streaming text-to-speech.
- Significantly reduced LLM operating cost by redesigning the conversation turn loop to remove redundant model calls from every turn.

- Built an automated evaluation harness that replays full interviews in seconds, with MLflow and LangSmith tracing to catch regressions across releases.

Tech: Python, FastAPI, LangGraph, Deepgram STT and TTS, WebRTC VAD, asyncio, WebSockets, Redis, MLflow, LangSmith, multi-provider LLMs

KEY PROJECTS

Self-Corrective RAG System (LangGraph, AWS Bedrock, FastAPI, MLflow, Docker, MCP) Live Demo

- Reduced hallucinations by **40%** and reached **92%** accuracy with a self-correcting multi-agent pipeline (document grading, hallucination checks, query rewriting) on Llama 3.1 70B via AWS Bedrock, with **sub-1.5s** responses.

End-to-End Litter Detection System (YOLOv8, PyTorch, OpenCV, CVAT, DVC, MLflow, Streamlit) Live Demo

- Improved small-object detection accuracy (mAP) by **15%** on **1,500+** hand-annotated images across **50+** tracked experiments, deployed as a live Streamlit app on a reproducible DVC and MLflow pipeline.

Character-Level Language Model from Scratch (PyTorch, NumPy, custom autograd) GitHub

- Implemented a character-level language model and a custom automatic differentiation engine from first principles to understand how modern LLMs train and generate text.

TECHNICAL SKILLS

AI / LLM: LLM Agents, Tool Use and Structured Output, RAG, Embeddings (Voyage AI), pgvector, LangGraph, LangChain, MCP, Anthropic Claude API, Hugging Face Transformers, PEFT/LoRA, Prompt Engineering, NLP, Computer Vision

Real-Time AI: WebSockets, asyncio, Deepgram STT and TTS, WebRTC VAD, Web Audio API, Streaming Pipelines

Backend: Python 3.12, FastAPI, SQLAlchemy, Alembic, PostgreSQL, MySQL, Redis, Node.js, Express, REST APIs

ML and Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, YOLOv8, OpenCV

DevOps and MLOps: Docker, Kubernetes, GitHub Actions (OIDC), CI/CD, pytest, ruff, Linux, Bash, MLflow, LangSmith, DVC, Model Monitoring

Cloud: AWS (App Runner, ECS Fargate, RDS, S3, SQS, Secrets Manager, Bedrock, EC2), Microsoft Azure, Google Cloud, BigQuery

Frontend and Integrations: JavaScript, HTML, CSS, Tailwind, Streamlit, WhatsApp Cloud API and Flows, Supabase Auth, Resend

Languages: English (Fluent), Malayalam (Native), Hindi (Elementary), Arabic (Elementary)

EDUCATION

Bachelor of Computer Applications (BCA), Yenepoya University, India **Aug 2022 to Nov 2025**

Specialisation in Machine Learning, AI and Robotics; industry-partnered programme with TCS and Microsoft

CERTIFICATIONS

- **Microsoft Azure:** AI Fundamentals (AI-900), Azure Fundamentals (AZ-900), Azure Data Fundamentals (DP-900)
- **IBM (Coursera), Generative AI:** Advanced LLM Fine-Tuning, Fine-Tuning Transformers, AI Agents with RAG and LangChain, LLM Architecture and Data Preparation, Language Models with Transformers, Foundational Models for NLP
- **IBM (Coursera):** Exploratory Data Analysis for Machine Learning
- **Kaggle:** Advanced SQL, Intro to SQL